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A Higher Thigh Muscle Injury Incidence in Professional Male Soccer Players Returning to Play After Anterior Cruciate Ligament Reconstruction

Analysis of 110 Cases From the UEFA Elite Club Injury Study

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Background: Previous literature has suggested an important association between a history of anterior cruciate ligament (ACL) reconstruction (ACLR) and future thigh muscle injuries, but more and larger studies are needed, particularly for quadriceps muscle injuries.

Purpose: To determine the incidence of hamstring and quadriceps muscle injuries after return to play (RTP) from ACLR using patellar tendon (PT) and hamstring tendon (HT) autografts in professional male soccer players.

Study Design: Cohort study; Level of evidence, 3.

Methods: A total of 5447 players from 74 clubs were followed prospectively from July 2001 to May 2022. The injury incidence (number of injuries per 1000 hours) was compared between players with ACLR and those without ACL injuries as well as for players with ACLR between a 2-year period before the ACL injury and a 2-year period after RTP. The rate ratio (RR) was calculated with the 95% confidence interval (CI) and examined for significance with *Z* statistics.

Results: There were 110 ACLR cases (76 PT and 34 HT grafts) in 107 players included. Players with ACLR had a 60% higher thigh muscle injury incidence (RR, 1.6 [95% CI, 1.3-2.0]) within 2 years after RTP compared with players without ACL injuries. Players with ACLR had a doubled thigh muscle injury incidence (RR, 2.0 [95% CI, 1.4-2.9]) in the 2 years after RTP compared with 2 years before their ACL injury. Players with a PT autograft had a higher quadriceps muscle injury incidence (RR, 3.5 [95% CI, 1.4-8.8]) in the 2 years after RTP compared with 2 years before their ACL injury, whereas there was no difference in the hamstring muscle injury incidence in players with an HT autograft (RR, 1.7 [95% CI, 0.5-5.8]).

Conclusion: ACLR in professional male soccer players was associated with an increased thigh muscle injury incidence, particularly quadriceps muscle injuries in players with a PT autograft.

Keywords: soccer; quadriceps muscle injury; ACL reconstruction; autograft

Anterior cruciate ligament (ACL) and thigh muscle injuries have been reported as the 2 most burdensome injuries in professional male soccer players.[3] ACL injuries are infrequent but are among the most severe injuries, with a mean absence of 6 to 10 months.[5,9,13,15,19,26,28,29,31] On the other hand, thigh muscle injuries, and particularly hamstring muscle injuries, are among the most frequent problems for soccer players.[7]

There are a few previous reports on the potential association between a history of ACL injuries and subsequent hamstring injuries. A tendency toward a higher proportion of hamstring injuries was found in Australian professional soccer players with a history of ACL injuries.[24] A more recent study from the same research group, extending the sample to include professional Australian Football League players and elite/subelite rugby union players, confirmed this finding, with approximately double the odds for a future hamstring injury with a history of ACL reconstruction (ACLR).[16] However, the link between previous ACLR and thigh muscle injuries has not been examined extensively, particularly in relation to quadriceps muscle

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injuries. It is also unclear whether there are differences between various grafts for ACLR in terms of increased predisposition to hamstring and quadriceps muscle injuries.

As such, the primary purpose of this study was to investigate the rate of hamstring and quadriceps injuries after return to play (RTP) from ACLR in professional male soccer players. Specifically, players with ACLR were compared with players without ACL injuries and to themselves before their ACL injury. The secondary purpose was to determine the influence of patellar tendon (PT) and hamstring tendon (HT) autografts as a risk factor for future hamstring or quadriceps muscle injuries after ACLR.

METHODS

This was a substudy of an injury surveillance study on professional male soccer players, the Elite Club Injury Study (ECIS), which has been carried out in collaboration with the UEFA (Union of European Football Associations) since 2001.[27] In total, 74 clubs with 485 team-seasons were followed prospectively over a variable number of seasons per team from July 2001 to May 2022. Details on the study design and data collection have been reported previously,[10] and the methodology adheres to recent consensus statements for injury surveillance.[4,30]

Inclusion and Exclusion Criteria

All players with a first team contract were eligible for inclusion. In this substudy, players with a complete ACL rupture undergoing ACLR using a PT or HT autograft were included because these 2 grafts comprise a vast majority of ACLR cases in this setting.[6,28] Players with bilateral ACLR procedures were included, but players with ACL injuries treated nonoperatively, revision ACLR, multiligament reconstruction including ACLR, and ACLR using allografts or other autografts than PT or HT grafts were excluded. All remaining players in the ECIS without ACL injuries served as external controls.

Definitions and Data Collection

A designated medical staff member from each club was responsible for sending exposure report forms and general injury report forms to the research center monthly.[10] Training and match exposures were recorded individually in minutes for each activity. An injury was defined using a time-loss definition as "any physical complaint sustained by a player that resulted from a football match or football training and led to the player being unable to take full part in future football training or match play."[4,10,30] For every ACL injury reported, an ACL injury–specific injury report form was also completed that requested information on the extent of the rupture, potential associated injuries, surgery date, and graft choice. The RTP date was identified on the exposure report form based on when players started attending team training sessions after ACLR or via the club medical staff if the player was no longer on the roster of a team in the study (ie, team relegated from the highest league system, player on loan or transferred to another club, etc). For every thigh muscle injury reported, details were requested to encompass whether it was a hamstring or quadriceps muscle injury, but the specific thigh muscle within these 2 muscle groups was not necessarily reported (being optional free-text information on the general injury report form).

Statistical Analysis

Data were analyzed using SPSS Statistics for Windows (Version 26.0; IBM). Kaplan-Meier survival curves were used to illustrate potential differences in the training and match exposure times to the first thigh muscle injury between players with ACLR and players without ACL injuries. The injury incidence was calculated as the number of injuries per 1000 hours ((Σ injuries/Σ exposure hours) × 1000) with the corresponding 95% confidence interval (CI). The rate ratio (RR) was calculated with the 95% CI and examined for significance with Z statistics. Between-group comparisons of the injury incidence, regardless of the type[7,25] or severity,[4,10,30] were made between players with ACLR and players without ACL injuries, serving as external controls. For calculation of the injury incidence, all exposures and injuries for the players without ACL injuries were included, while only the first and second years before the ACL injury as well as the first and second years after RTP in players with ACLR were included. Subanalyses were conducted for moderate and severe injuries (>7 days of absence) because most structural muscle

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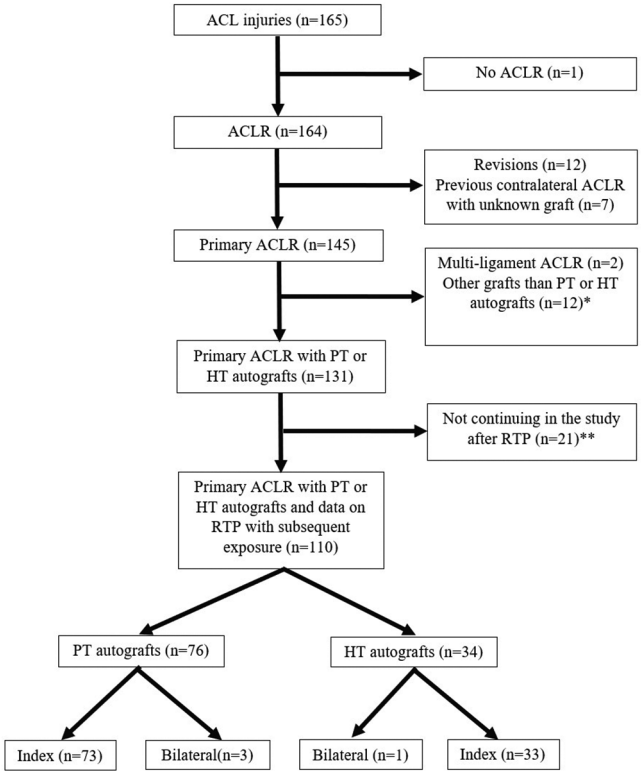


Figure 1. Flowchart of included players with anterior cruciate ligament reconstruction (ACLR).

injuries fall within this interval.[7,8] Within-group comparisons of the injury incidence were made for players with ACLR before their ACL injury, serving as internal controls, and after RTP. The primary time periods assessed were 2 years before the ACL injury and 2 years after RTP. Subanalyses were conducted identically as above for 1-year time periods.

RESULTS

A total of 3,266,563 exposure hours were registered (2,762,227 training hours and 504,336 match hours) by 5447 unique players over the 21-season study period. During this period, there were 126 players who met the inclusion criteria and had suffered 131 ACL injuries (5 players suffered a contralateral ACL injury). The injured players did not continue their participation in the ECIS after RTP (because of loans, transfers, clubs not continuing their participation, etc) in 21 cases, leaving 110 ACL injuries for analysis (Figure 1).

Players With ACLR Versus Players Without ACL Injuries

Among the 110 ACLR cases, 33 (30%) and 43 (39%) suffered 56 and 80 thigh muscle injuries within 1 and 2 years after RTP, respectively. In comparison, there were 5238

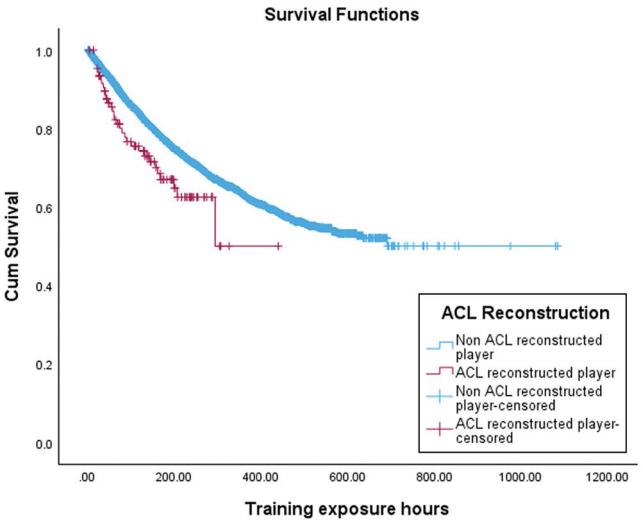


Figure 2. Training hours before a thigh muscle injury in players with anterior cruciate ligament (ACL) reconstruction (ACLR) after return to play (red) and in players without ACL injuries (blue).

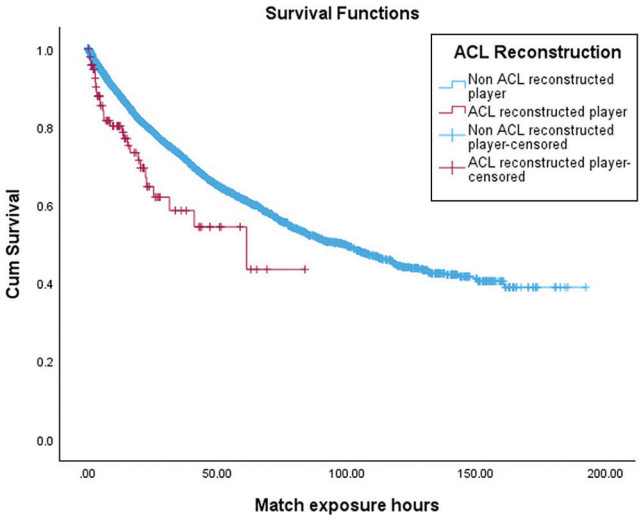


Figure 3. Match hours before a thigh muscle injury in players with anterior cruciate ligament (ACL) reconstruction (ACLR) after return to play (red) and in players without ACL injuries (blue).

players without ACL injuries, and 1211 (23%) and 1479 (28%) of these players suffered 1665 and 2372 thigh muscle injuries during the first 1- and 2-year periods in the study, respectively. Between-group differences in the number of training and match hours until the first thigh muscle injury are shown in Figures 2 and 3.

Players with ACLR had a 40% to 150% higher overall thigh muscle injury incidence (RR, 1.6), overall quadriceps muscle injury incidence (RR, 2.5), and overall hamstring muscle injury incidence (RR, 1.4) up to 2 years after RTP

TABLE 1
Injury Incidence and RR of Players With ACLR After RTP Versus Players Without ACL Injuries[a]

	Injury Incidence (95% CI)			Injury Incidence (95% CI)	
	Without ACL Injuries	With ACLR at ≤1 y After RTP	RR (95% CI)	With ACLR at ≤2 y After RTP	RR (95% CI)
Thigh muscle injuries					
Training	0.8 (0.8-0.8)	1.7 (1.2-2.4)	2.2 (1.5-3.1)	1.7 (1.2-2.2)	2.1 (1.5-2.8)
Match	6.3 (6.0-6.5)	10.6 (7.1-15.9)	1.7 (1.1-2.5)	9.3 (6.7-12.8)	1.5 (1.1-2.1)
Total	1.6 (1.6-1.7)	2.7 (2.1-3.5)	1.7 (1.3-2.2)	2.6 (2.1-3.3)	1.6 (1.3-2.0)
Hamstring muscle injuries					
Training	0.5 (0.5-0.5)	0.8 (0.4-1.3)	1.5 (0.9-2.6)	0.8 (0.5-1.3)	1.7 (1.1-2.6)
Match	4.9 (4.7-5.2)	7.1 (4.3-11.6)	1.4 (0.9-2.3)	6.9 (4.8-10.1)	1.4 (1.0-2.1)
Total	1.2 (1.1-1.2)	1.4 (1.0-2.1)	1.2 (0.9-1.8)	1.6 (1.2-2.1)	1.4 (1.0-1.8)
Quadriceps muscle injuries					
Training	0.3 (0.2-0.3)	0.9 (0.6-1.5)	3.4 (2.1-5.5)	0.8 (0.5-1.2)	2.9 (1.9-4.5)
Match	1.1 (1.0-1.2)	3.5 (1.8-7.1)	3.2 (1.6-6.4)	2.3 (1.2-4.5)	2.1 (1.1-4.0)
Total	0.4 (0.4-0.4)	1.2 (0.8-1.8)	3.0 (2.0-4.5)	1.0 (0.7-1.4)	2.5 (1.7-3.5)
Moderate/severe thigh muscle injuries					
Training	0.5 (0.4-0.5)	1.5 (1.0-2.1)	3.1 (2.1-4.6)	1.3 (0.9-1.8)	2.8 (2.0-3.9)
Match	4.7 (4.5-4.9)	8.9 (5.7-13.7)	1.9 (1.2-2.9)	9.3 (6.7-12.8)	2.0 (1.4-2.7)
Total	1.1 (1.1-1.2)	2.3 (1.7-3.0)	2.0 (1.5-2.7)	2.3 (1.8-2.9)	2.1 (1.6-2.6)
Moderate/severe hamstring muscle injuries					
Training	0.3 (0.3-0.3)	0.6 (0.3-1.1)	2.1 (1.1-3.8)	0.6 (0.4-1.0)	2.1 (1.3-3.5)
Match	3.8 (3.7-4.0)	5.8 (3.3-9.9)	1.5 (0.9-2.6)	6.9 (4.8-10.1)	1.8 (1.2-2.7)
Total	0.8 (0.8-0.9)	1.2 (0.8-1.7)	1.4 (0.9-2.1)	1.4 (1.0-1.9)	1.7 (1.3-2.3)
Moderate/severe quadriceps muscle injuries					
Training	0.2 (0.2-0.2)	0.8 (0.5-1.3)	4.7 (2.8-7.9)	0.7 (0.4-1.1)	3.9 (2.5-6.3)
Match	0.8 (0.7-0.9)	3.1 (1.5-6.5)	3.8 (1.8-7.9)	2.3 (1.2-4.5)	2.8 (1.4-5.4)
Total	0.3 (0.3-0.3)	1.1 (0.7-1.6)	3.9 (2.6-6.0)	0.9 (0.6-1.3)	3.3 (2.2-4.8)

[a]Injury incidence is expressed as the number of injuries per 1000 hours. RR is expressed as players with ACLR divided by players without ACL injuries. ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; RR, rate ratio; RTP, return to play.

compared with players without ACL injuries (Table 1). Similar findings were seen up to 1 year after RTP and for moderate/severe muscle injuries, but with larger RRs except for overall hamstring muscle injuries. Time to return to training after ACLR was not significantly associated with the overall thigh muscle injury incidence. Players who had returned to training within 6 months had a nonsignificant 30% (RR, 0.7; $P = .20$) and 40% (RR, 0.6; $P = .15$) lower thigh muscle injury incidence compared with those who had returned to training at 6 to 9 months and >9 months, respectively.

Players With ACLR Before ACL Injury Versus After RTP

Players with ACLR had a significantly higher thigh muscle injury incidence during training, matches, and overall in the first year after RTP (n = 31, n = 24, and n = 55, respectively) compared with the year before their ACL injury (n = 11, n = 17, and n = 28, respectively) (Table 2). This pattern was also seen in the 2 years after RTP (n = 43, n = 36, and n = 79, respectively) compared with the 2 years before the ACL injury (n = 13, n = 36, and n = 49, respectively), but with no statistical significance for match injuries.

Players with a PT autograft had a higher thigh muscle injury incidence after ACLR, regardless of whether 1 or 2 years were compared before the ACL injury and after RTP (Table 3). Particularly, an increased incidence was seen for quadriceps muscle injuries, with RRs of 4.1 and 3.5, respectively.

Conversely, although players with an HT autograft had a higher thigh muscle injury incidence after ACLR for the 1-year comparison before the ACL injury and after RTP, there were no significant differences in quadriceps and hamstring muscle injury incidences for players with an HT autograft (Table 4).

DISCUSSION

The main finding of this study was a noteworthy increase in the thigh muscle injury incidence among professional male soccer players after RTP from ACLR. This pattern was seen compared with players without ACL injuries (external control) and to the players' thigh muscle injury incidence during a 2-year time frame before their ACL injury (internal control).

Future Thigh Muscle Injuries

Thigh muscle injuries are recognized as common occurrences in soccer, leading to substantial time loss and a high

TABLE 2
Injury Incidence and RR of Players With ACLR After RTP Versus Before ACL Injury[a]

	Injury Incidence (95% CI)		
	≤1 y Before ACL Injury	≤1 y After RTP	RR (95% CI)
Thigh muscle injuries			
Training	0.5 (0.3-1.0)	1.7 (1.2-2.4)	3.3 (1.6-6.6)
Match	4.3 (2.6-7.1)	10.6 (7.1-15.9)	2.5 (1.3-4.7)
Total	1.1 (0.8-1.7)	2.7 (2.1-3.5)	2.4 (1.5-3.9)
Hamstring muscle injuries			
Training	0.1 (0.0-0.4)	0.8 (0.4-1.3)	7.1 (1.6-31.4)
Match	3.4 (1.9-6.0)	7.1 (4.3-11.6)	2.1 (1.0-4.4)
Total	0.6 (0.4-1.1)	1.4 (1.0-2.1)	2.3 (1.2-4.4)
Quadriceps muscle injuries			
Training	0.3 (0.1-0.7)	0.9 (0.6-1.5)	2.9 (1.1-7.3)
Match	0.9 (0.3-2.7)	3.5 (1.8-7.1)	4.1 (1.1-15.6)
Total	0.4 (0.2-0.8)	1.2 (0.8-1.8)	3.0 (1.4-6.4)
	Injury Incidence (95% CI)		
	≤2 y Before ACL Injury	≤2 y After RTP	RR (95% CI)
Thigh muscle injuries			
Training	0.4 (0.2-0.7)	1.7 (1.2-2.2)	4.5 (2.3-9.0)
Match	6.3 (4.5-8.8)	9.3 (6.7-12.8)	1.5 (0.9-2.4)
Total	1.3 (1.0-1.8)	2.6 (2.1-3.3)	2.0 (1.4-2.9)
Hamstring muscle injuries			
Training	0.1 (0.0-0.3)	0.8 (0.5-1.3)	11.3 (2.7-48.3)
Match	5.1 (3.5-7.5)	6.9 (4.8-10.1)	1.3 (0.8-2.3)
Total	0.9 (0.6-1.3)	1.6 (1.2-2.1)	1.8 (1.1-2.9)
Quadriceps muscle injuries			
Training	0.2 (0.1-0.5)	0.8 (0.5-1.2)	3.6 (1.5-8.9)
Match	1.0 (0.4-2.3)	2.3 (1.2-4.5)	2.4 (0.8-7.2)
Total	0.3 (0.2-0.6)	1.0 (0.7-1.4)	2.9 (1.5-5.8)

[a]Injury incidence is expressed as the number of injuries per 1000 hours. RR is expressed as after ACLR divided by before ACL injury. ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; RR, rate ratio; RTP, return to play.

TABLE 3
Injury Incidence and RR of Players With ACLR Using Patellar Tendon Autograft After RTP Versus Before ACL Injury[a]

	Injury Incidence (95% CI)		
	≤1 y Before ACL Injury	≤1 y After RTP	RR (95% CI)
Thigh muscle injuries			
Training	0.6 (0.3-1.2)	1.8 (1.2-2.7)	3.2 (1.4-7.6)
Match	4.4 (2.4-7.9)	12.0 (7.7-18.8)	2.8 (1.3-5.8)
Total	1.2 (0.7-1.9)	2.9 (2.2-4.0)	2.5 (1.4-4.3)
Hamstring muscle injuries			
Training	0.2 (0.0-0.6)	1.0 (0.6-1.7)	6.2 (1.4-27.5)
Match	4.0 (2.1-7.4)	8.2 (4.8-14.1)	2.1 (0.9-4.7)
Total	0.8 (0.4-1.4)	1.8 (1.2-2.7)	2.3 (1.1-4.5)
Quadriceps muscle injuries			
Training	0.2 (0.1-0.7)	0.7 (0.4-1.4)	3.1 (0.8-11.4)
Match	0.4 (0.1-2.8)	3.8 (1.7-8.4)	9.6 (1.2-79.5)
Total	0.3 (0.1-0.7)	1.1 (0.6-1.8)	4.1 (1.4-12.3)
	Injury Incidence (95% CI)		
	≤2 y Before ACL Injury	≤2 y After RTP	RR (95% CI)
Thigh muscle injuries			
Training	0.4 (0.2-0.7)	1.7 (1.2-2.4)	4.8 (2.1-10.8)
Match	6.8 (4.7-9.9)	10.6 (7.5-15.1)	1.6 (0.9-2.6)
Total	1.4 (1.0-2.0)	2.9 (2.3-3.7)	2.0 (1.3-3.1)
Hamstring muscle injuries			
Training	0.1 (0.0-0.4)	1.0 (0.6-1.6)	9.7 (2.2-41.7)
Match	5.8 (3.8-8.7)	8.2 (5.5-12.3)	1.4 (0.8-2.5)
Total	1.1 (0.7-1.6)	2.0 (1.5-2.7)	1.9 (1.1-3.1)
Quadriceps muscle injuries			
Training	0.2 (0.0-0.5)	0.7 (0.4-1.1)	4.3 (1.2-15.2)
Match	0.8 (0.2-2.3)	2.4 (1.1-5.0)	3.2 (0.8-12.3)
Total	0.3 (0.1-0.6)	0.9 (0.6-1.4)	3.5 (1.4-8.8)

[a]Injury incidence is expressed as the number of injuries per 1000 hours. RR is expressed as after ACLR divided by before ACL injury. ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; RR, rate ratio; RTP, return to play.

injury burden.[7,8] There is a paucity of specific studies on male professional soccer players exclusively, but an association between previous ACL injuries or ACLR and future hamstring muscle injuries has been reported in various high-level athletes, in particular Australian rules footballers.[12,16,20,21,24] Our study corroborates these findings in which previous ACLR increased the hamstring muscle injury incidence compared with players without ACL injuries. Interestingly, both PT and HT grafts seemed to predispose athletes to future hamstring reinjuries in one of these studies,[12] which is also partly in line with our study, showing that a PT graft was associated with a higher hamstring muscle injury incidence. In addition, prior ACLR, in particular using a PT graft, increased the quadriceps muscle injury incidence significantly compared with before the ACL injury, which is a novel finding that has not been previously described in the literature. Quadriceps inhibition and quadriceps strength deficits have historically been common after ACLR in professional male football players with either HT or PT grafts[11]; this might be a risk factor for subsequent quadriceps muscle injuries after ACLR but needs to be addressed in future studies.

Consequences of ACL Injuries

Importantly, an ACL injury not only affects the knee joint itself but can also lead to global functional impairments including the brain[2,18] and muscles of the lower extremities,[1] which possibly may influence the subsequent injury risk. It is well documented that after ACLR, patients often report deficits in quadriceps muscle strength, knee proprioception, and lower limb neuromuscular control that can persist for years.[17] Partial joint deafferentation can lead to compensatory movement patterns of the lower limb in an attempt of underloading the knee joint.[22] These intralimb and interlimb compensations are likely connected to subtle adjustments in kinematics and kinetics of the whole lower limb during common soccer movements, such as high-speed running or deceleration. In the presence of concomitant muscle strength deficits, these changes may predispose athletes to further compensatory injuries affecting the muscles of the lower limb.

There are several graft choices for ACLR in which PT and HT autografts have been the most widely adopted in professional players during recent decades.[5,28] Regardless

TABLE 4
Injury Incidence and RR of Players With ACLR Using Hamstring Tendon Autograft After RTP Versus Before ACL Injury[a]

	Injury Incidence (95% CI)		
	≤1 y Before ACL Injury	≤1 y After RTP	RR (95% CI)
Thigh muscle injuries			
Training	0.5 (0.2-1.5)	1.6 (0.9-3.0)	3.3 (0.9-12.1)
Match	4.1 (1.5-10.8)	7.4 (3.1-17.8)	1.8 (0.5-6.8)
Total	1.0 (0.5-2.1)	2.2 (1.3-3.7)	2.2 (0.9-5.5)
Hamstring muscle injuries			
Training	0.0 (0.0-0.0)	0.3 (0.1-1.3)	NA
Match	2.0 (0.5-8.1)	4.4 (1.4-13.8)	2.2 (0.4-13.1)
Total	0.3 (0.1-1.1)	0.7 (0.3-1.8)	2.6 (0.5-13.5)
Quadriceps muscle injuries			
Training	0.5 (0.2-1.5)	1.3 (0.7-2.6)	2.7 (0.7-10.1)
Match	2.0 (0.5-8.1)	3.0 (0.7-11.8)	1.5 (0.2-10.3)
Total	0.7 (0.3-1.7)	1.5 (0.8-2.7)	2.1 (0.7-6.1)
	Injury Incidence (95% CI)		
	≤2 y Before ACL Injury	≤2 y After RTP	RR (95% CI)
Thigh muscle injuries			
Training	0.4 (0.1-1.2)	1.6 (0.9-2.7)	4.1 (1.2-14.2)
Match	4.7 (2.1-10.5)	5.2 (2.2-12.4)	1.1 (0.3-3.6)
Total	1.0 (0.5-1.9)	2.0 (1.2-3.1)	2.0 (0.9-4.3)
Hamstring muscle injuries			
Training	0.0 (0.0-0.0)	0.5 (0.2-1.3)	NA
Match	3.1 (1.2-8.4)	3.1 (1.0-9.6)	1.0 (0.2-4.4)
Total	0.4 (0.2-1.2)	0.8 (0.4-1.6)	1.7 (0.5-5.8)
Quadriceps muscle injuries			
Training	0.4 (0.1-1.2)	1.1 (0.6-2.1)	2.8 (0.8-10.4)
Match	1.6 (0.4-6.3)	2.1 (0.5-8.3)	1.3 (0.2-9.4)
Total	0.6 (0.2-1.3)	1.2 (0.7-2.2)	2.1 (0.7-6.2)

[a]Injury incidence is expressed as the number of injuries per 1000 hours. RR is expressed as after ACLR divided by before ACL injury. ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; NA, not assessed; RR, rate ratio; RTP, return to play.

of the autograft choice, graft harvest is associated with donor site morbidity and corresponding muscle weakness, which possibly could lead to an increased risk of sustaining thigh muscle injuries. It is well documented that approximately 8 of 10 hamstring muscle injuries in professional male soccer players are located at the biceps femoris and not the medial hamstring.[7,8] Interestingly, therefore, it has been reported that the biceps femoris compensates for semitendinosus function in male soccer players with ACLR using HT autografts, as evaluated by functional magnetic resonance imaging,[23] but the influence on the thigh muscle and knee injury risk is yet to be determined. Despite that there was no significant effect of the harvest and use of HT autografts on subsequent hamstring muscle injuries, our results (RR, 1.7) warrant a better understanding of graft selection and its potential effect on muscle function and the injury risk after ACLR. From our data, the use of an HT autograft was associated with a lower hamstring muscle injury incidence before ACLR compared with the use of a PT autograft, and this might be a key criterion among surgeons for using HT autografts in some players, but this also needs further research.

Future Perspectives

Moving forward, there is a need for further research to elucidate the underlying mechanisms driving the observed increase in the thigh muscle injury rate after ACLR in professional male soccer players. Longitudinal studies incorporating comprehensive biomechanical assessments, muscle imaging techniques, functional performance measures, and load monitoring are warranted to better understand the mechanisms contributing to the altered injury risk in this population. This in turn may help create targeted rehabilitation interventions that are tailored to address specific deficits and risk factors associated with different autograft types to mitigate the higher susceptibility of thigh muscle injuries in soccer players after ACLR.

Limitations

This study has some limitations. First, it was impossible to create a "pure" control group without a previous thigh muscle injury or a previous ACL injury because we had no access to baseline data for all previous injuries. Second, more than two-thirds of the cases had a PT autograft, which might introduce a sparse-data bias for HT autografts, leading to a type II error for the RR.[14] Third, our findings cannot be generalizable to other grafts, such as the quadriceps tendon autograft or various allografts, which were rarely used in our cohort. Fourth, the sample did not allow for subanalyses between structural and functional muscle injuries,[7,25] but instead, a subanalysis was conducted for moderate and severe muscle injuries (>7 days of absence), which included relatively few functional muscle injuries, showing the same trend as the main analysis. Fifth, because it was only optional to specify the individual muscle with the muscle group on the general injury report form, the sample did not allow us to analyze, for example, the semitendinosus muscle exclusively. Sixth, we did not analyze whether a player with less than full match exposure started the match or came on from the bench, which possibly could influence the thigh muscle injury incidence because the proportion of hamstring muscle injuries is known to increase toward the end of both match halves.[8] Finally, the lack of universal RTP criteria, with objective data from physical and psychological testing before RTP, could potentially influence the comparison of muscle injury incidences between groups.

CONCLUSION

There was an increased thigh muscle injury incidence in the 2 years after RTP in professional male soccer players who underwent ACLR compared with the same time frame before their ACL injury and compared with players without ACL injuries. Specifically, players with a PT autograft had a 4- and 3-fold higher quadriceps muscle injury incidence at 1 and 2 years after RTP compared with before the ACL injury, respectively.

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